

RAKESH SHANKAR

AI ENGINEER

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SUMMARY

A passionate AI and Machine Learning enthusiast with hands-on experience in deep learning, spatial reasoning, and cyber threat intelligence. Skilled in building intelligent systems, from predictive analytics to adaptive testing models. Strong research acumen with expertise in data-driven decision-making, automation, and real-time reasoning. Adept at integrating AI with real-world applications.

TECHNICAL SKILLS

Python	Computer Vision	Deep Learning
Numpy	Transformers	AutoML
SQL	NLP	Data Analysis
Pandas	Scikit-Learn	Pytorch, Tensorflow

PROFESSIONAL EXPERIENCE

Karunya Hacks AI Club- Trainer **Aug 2023 - Jan 2024**

- Actively engaging in the learning process, exploring machine learning, deep learning, and real-world AI applications through interactive discussions and hands-on sessions.

PROJECTS

ECG Anomaly Detection Using LSTM Autoencoders

- Accomplished accurate detection of abnormal ECG patterns as measured by high anomaly detection precision and recall by training an LSTM autoencoder to learn normal heart rhythms and flag deviations as anomalies.

AI-powered-Cyber-Threat-Reasoning

- Achieved automated cyber threat analysis and intelligent decision-making by leveraging the Gemini API to process threat intelligence data, resulting in concise summaries and actionable insights for improved security response.

Directory Describing System

- Achieved precise summarization of directory contents using a Python tool with the Gemini API, processing over 30 files (.docx, .pdf, .csv) and subfolders effectively. Extracted metadata (e.g., size, type) and snippets (up to 500 characters) with PyPDF2 and python-docx, caching in JSON for efficiency. Improved organization with human-like descriptions and reasoning, enhancing folder structure and documentation.

Brain Tumor Segmentation

- Developed a deep learning model for brain tumor segmentation using a U-Net architecture with a VGG19 encoder. The dataset was preprocessed to extract MRI images and corresponding tumor masks. Implemented a custom data generator for efficient loading and augmentation. Used TensorFlow and Keras to build and train the model with GPU acceleration, optimizing performance with techniques like learning rate scheduling, early stopping, and mixed-precision training.

EDUCATION

B.Tech CSE(Artificial Intelligence and Machine Learning) **Sept 2022 - May 2026(present)**

Karunya University

- Major in Artificial Intelligence and Machine Learning

High School

- SSVM World School, Coimbatore

ADDITIONAL INFORMATION

- **Languages:** Tamil, English
- **Certifications:** Feature Engineering, Pandas, Introduction to Machine Learning
- **Awards/Activities:** Smart India Hackathon 2023 Finalist